



DEPARTMENT OF COMPUTER SCIENCE
UNIVERSITY OF ENGINEERING & TECHNOLOGY,
PESHAWAR

Course Title: Web Technologies
Course Code: CS 224
Instructor: Mr. Mohammad

Total Marks: 50 Marks

Submission Guidelines

- Submit complete files on the assignment on Google Classroom.
- Upload PDF report of the assignment.
- Upload video with voice explaining the implementation of each task.
- Submit the GitHub repository link and the live website URL in Google Classroom private comment section.
- Use the provided title page for your assignment.

Assignment No. 02

Online Bookstore Management API

You have been hired by a small online bookstore to develop a backend system. The system should allow the store to manage its books: view, add, update, and delete books. The system should be built as a RESTful API using Node.js, Express, and MongoDB.

Your task is to develop the backend API according to the requirements below and make it accessible locally or on a cloud platform like MongoDB Atlas.

Requirements:

1. Environment Setup (10 points)

- Install **Node.js** and **npm**
- Install required packages:
 - express
 - mongoose
 - body-parser (optional)
 - dotenv for environment variables
 - nodemon (optional, for development)

- Connect to **MongoDB** (local or Atlas cloud instance)

2. MongoDB Schema (10 points)

- Create a **Book schema** with the following fields:
 - title (string, required)
 - author (string, required)
 - genre (string)
 - price (number, required)
 - publishedDate (date)
 - inStock (boolean, default: true)

3. API Routes (10 points)

Implement the following **RESTful API endpoints** using Express:

Endpoint	Method	Description
/api/books	GET	Get all books
/api/books/:id	GET	Get a single book by ID
/api/books	POST	Add a new book
/api/books/:id	PUT	Update an existing book by ID
/api/books/:id	DELETE	Delete a book by ID

Additional requirements:

- Use **JSON format** for request and response
- Validate required fields on POST and PUT
- Return **appropriate HTTP status codes**:
 - 200 → success
 - 201 → created
 - 400 → bad request
 - 404 → not found

4. Middleware & Error Handling (10 points)

- Implement a middleware to log each request (method, endpoint, date/time)
- Implement a global error handler to catch invalid routes or database errors
- Return clear error messages in JSON format

5. Additional Features (10 points)

- Add search functionality: GET /api/books?author=xyz&genre=abc
- Add pagination for large book lists

Submission Requirements:

1. A zip folder containing:
 - package.json
 - server.js (or app.js)
 - models/Book.js
 - routes/books.js
 - .env file (without exposing credentials)
2. Postman collection or documentation showing all API routes tested
3. If deployed online (on Vercel), include the live API URL **(05 Marks)**